Midterm 1 Answer Key (W2013)

1) (b)

$$P = 200 + \overset{.5083}{(P|F, 7\%, 10)}\Big[225\overset{13.8164}{(F|A, 7\%, 10)} + 25\overset{8.6540}{(F|A, 7\%, 7)} + 25\overset{5.7507}{(F|A, 7\%, 5)} + 25\overset{2.0700}{(F|A, 7\%, 2)}\Big]$$

$$= 1989.50$$

2) (d)

3) (e) $4^{\$} = 1^{\$}(F|P, 2\%, N)$

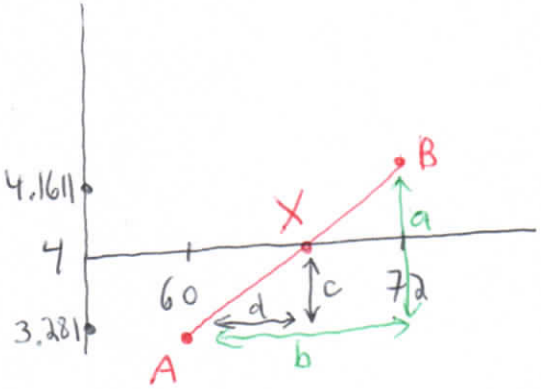
2 methods

$(F|P, 2\%, 60) = 3.2810$ A

$(F|P, 2\%, 72) = 4.1611$ B

or

$$\boxed{\begin{aligned} 4 &= (1.02)^N \\ \log 4 &= N \log 1.02 \\ N &= \frac{\log 4}{\log 1.02} = 70.006 \end{aligned}}$$

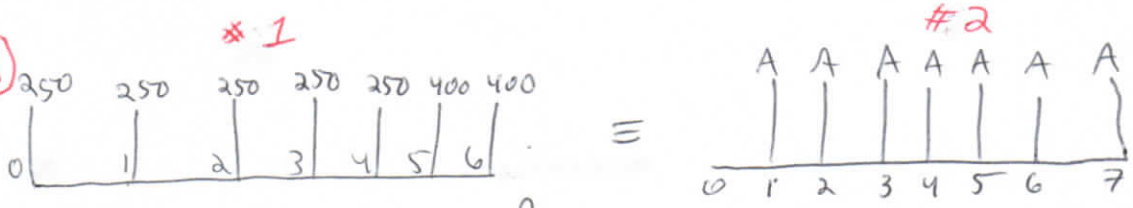


$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

$$(\overset{a}{4.1611 - 3.281})(\overset{d}{X - 60}) = (\overset{b}{72 - 60})(4 - 3.281)$$

$$X = 69.80 \approx 70$$

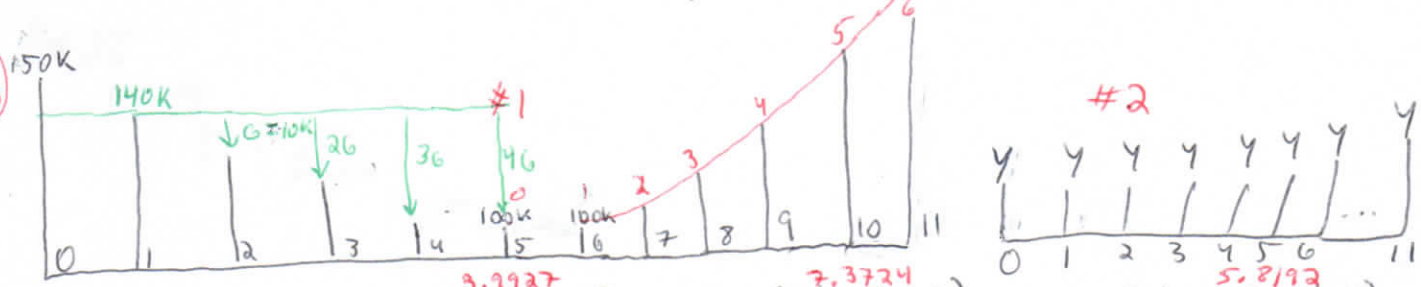
4) (d)



$$\overbrace{250 + 250\overset{4.9173}{(P|A, 6\%, 6)} + 150\overset{1.8334}{(P|A, 6\%, 2)}\overset{.7921}{(P|F, 6\%, 4)}}^{P_1} = \overbrace{A\overset{5.5824}{(P|A, 6\%, 7)}}^{P_2}$$

$$A = 304.02 \approx 304$$

5) (b)



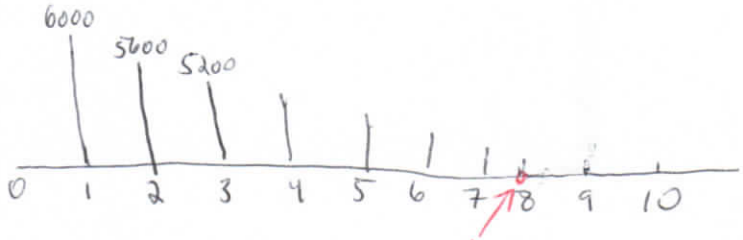
$$P_1 = 150{,}000 + 140{,}000\overset{3.9927}{(P|A, 8\%, 5)} - \underbrace{10{,}000\overset{7.3724}{(P|G, 8\%, 5)}}_{\text{linear gradient declining}} + \underbrace{100{,}000}_{A_1}\overset{5.8192}{(P|A, \underset{g}{10\%}, 8\%, 6)} \times \overset{.6806}{(P|F, 8\%, 5)}$$

$$P_2 = Y\overset{7.1390}{(P|A, 8\%, 11)} + Y$$

$P_1 = 1{,}031{,}308.752 = Y \times 7.1390 + Y = Y(7.1390+1)$

$$Y = \frac{1{,}031{,}308.752}{.1390} = 126{,}711.9735 \approx 126{,}712$$

6) a



V_8

$V_8 = 6000(F|A, 6\%, 8) - 400(P|G, 6\%, 8)(F|P, 6\%, 8)$ (9.8975, 19.8416, 1.5938)

$V_8 = 46{,}735.58$

7) e

$I_1 = 1600$ $\quad I_2 = 17274 \times .08 = 1381.92$ $\quad I_3 = 1146.39$

$PP_1 = 2726$ $\quad PP_2 = 4326 - 1381.92 = 2944.08$ $\quad PP_3 = 3179.61$

$B_1 = 17274$ $\quad B_2 = 17274 - 2944.08 = 14329.92$ $\quad B_3 = 11{,}150.31$

$I_4 = 892.03$

$PP_4 = 3433.97 \approx 3434$

OR

B_3 = Balance remaining at end of year 3 $= 4326(P|A, 8\%, 3) = 11{,}148.53$ (2.5771; 2 payments remaining)

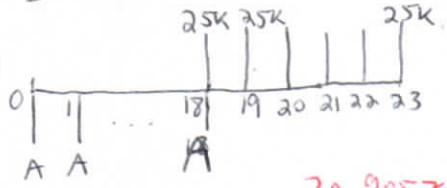
$I_4 = B_3 \times i = 4326(P|A, 8\%, 3) \times .08 = 891.88$

$A - I_4 = PP_4 = 4326 - 891.88 = 3434.12 \approx 3434$

OR

$B_3 - B_4 = 4326(P|A, 8\%, 3) - 4326(P|A, 8\%, 2) = 3433.98 \approx 3434$ (2.5771, 1.7833)

8) c



$A(F|P, 6\%, 18) + A(F|A, 6\%, 18) = 25{,}000 + 25000(P|A, 6\%, 5)$ (2.8543, 30.9057, 4.2124)

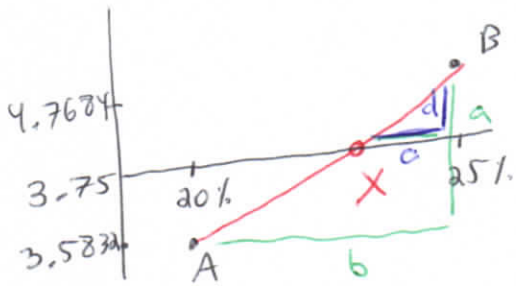
$$A = \frac{130{,}310}{33.76} = 3859.89$$

9) b $75{,}000 = 20{,}000(F|P, i, 7)$ OR $75000 = 20000(1+i)^7$

$3.75 = (F|P, i, 7)$ $\qquad 3.75 = (1+i)^7$

$(F|P, 20\%, 7) = 3.5832$ $\qquad (3.75)^{\frac{1}{7}} = 1+i$

$(F|P, 25\%, 7) = 4.7684$ $\qquad i = .2078 \approx 21\%$

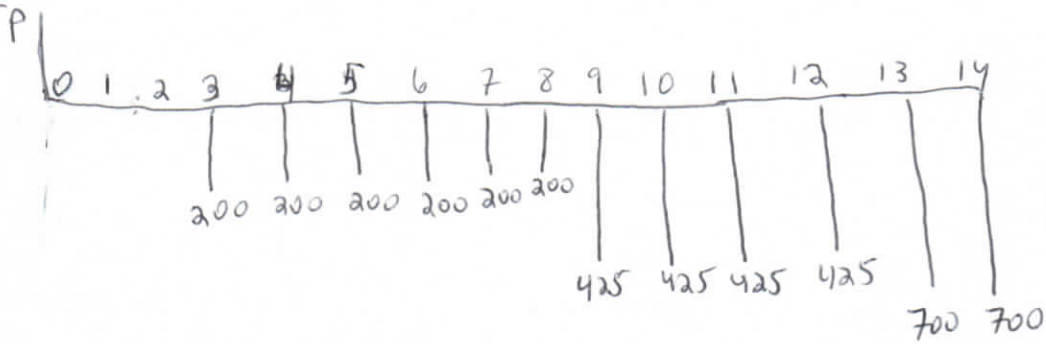


$$\frac{a}{b} = \frac{d}{c} \Rightarrow ac = db$$

$$(4.7684 - 3.5832)(25 - X) = (4.7684 - 3.75)(25 - 20)$$

$$X = 20.70 \approx 21\%$$

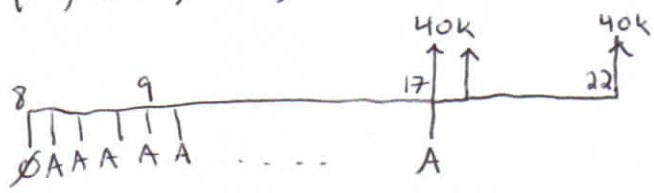
10) e $\left[200(F|A, 8\%, 12) + 225(F|A, 8\%, 6) + 275(F|A, 8\%, 2)\right](P|F, 8\%, 14)$



11) c) $i_m = \frac{6\%}{12} = .005$ or $.5\%$ monthly, $N = 12 \times 9$ years $= 108$.

$i_a = \left(1 + \frac{.06}{12}\right)^{12} - 1 = .06168$, $N = 6$ years $(-1$ year$)$.

$$A(\overset{142.7399}{F|A, .5\%, 108}) = 40{,}000 + 40{,}000(\overset{4.1932}{P|A, 6.168\%, 5})$$



$$A = \frac{40{,}000(1 + 4.1932)}{142.7399} = 1{,}455.29$$

12. c). $N = 12 \times 20 = 240$

$i_m = \frac{.06}{12} = .005$

$A = 200{,}000\ (A|P, .5\%, 240) = 1440.00$ (.0072)

$B_{60} = 1440\ (P|A, .5\%, 180) = 135{,}093.99$ (118.5035) (Balance after 5 years)

New monthly payments using $i_m = \frac{.09}{12} = .0075$

$A = 135{,}093.99\ (A|P, .75\%, 180) = 1369.85$ (.01014)

13. a) $2\cancel{P} = \cancel{P}(1+.01)^N$

$\log 2 = N \log 1.01$

$N = \frac{\log 2}{\log 1.01} = 69.6607$

\# of years $= \frac{69.6607}{12} = 5.8051 \approx 5.81$ years.

14. a) option 1: $1000\ (1+i_a)^{10} = 3000$

$(1+i_a)^{10} = 3$

$1 + i_a = 3^{\frac{1}{10}}$

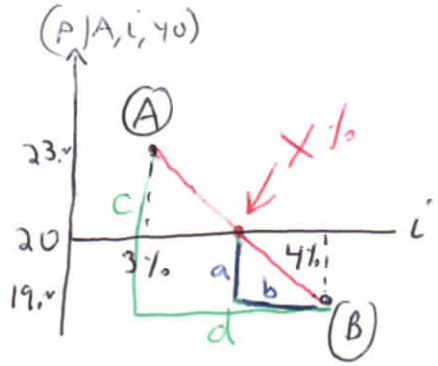
$i_a^{①} = 3^{\frac{1}{10}} - 1 = .1161 \approx 11.61\%$

option 2: $1000 = 50(P|A, i, 40)$

$20 = (P|A, i, 40)$

$(P|A, 3\%, 40) = 23.1148$ (A)

$(P|A, 4\%, 40) = 19.7928$ (B)



$\frac{a}{b} = \frac{c}{d}$

$ad = bc$

$(20 - 19.7928)(4 - 3\%) = (4 - X)(23.1148 - 19.7928)$

$\frac{.2072 \times 1\%}{3.322} = (4 - X)$

$X = 3.938\% = i_q \Rightarrow i_a^{②} = (1.03938)^4 - 1$

$= .1671 \approx 16.71\%$

$i_a^{②} - i_a^{①} = 5.1\%$

15) a) $P = 500{,}000(1 - .2) = 400{,}000$

1) $N = 20 \text{ yrs} \times 12 \text{ m} = 240 \text{ m}$.

APR = 6% quarterly $\Rightarrow i_m = \left(1 + \frac{.06}{4}\right)^{\frac{1}{3}} - 1 = .004975$

or .4975 %

$A_{①} = 400{,}000(A|P, \overset{.007147}{.4975\%}, 240) = 2858.80$

2) $N = 40 \times 12 = 480$.

APR = 8% $\quad i_m = \left(1 + \frac{.08}{4}\right)^{\frac{1}{3}} - 1 = .006623$

or .6623 %.

$A_{②} = 400{,}000\ (A|P, \overset{.006914}{.6623\%}, 480) = 2765.6$.

$A_{①} - A_{②} = 93.2$

16) e) $P = 125(P|A, \overset{9.3857}{4\%}, 12) + 5000(P|F, \overset{.6246}{4\%}, 12)$

$= 4296.1375$

17) b) $i_q = \frac{8\%}{4} = 2\%$ $\quad , N = 20 \times 4 = 80$

$A = 150{,}000\ (A|P, \overset{.0252}{2\%}, 80) = 3780$

18) d) coupon: \$100 annually, 4 payments of \$25 each.

current yield: $\frac{25\$}{800\$}$ per quarter or 3.125% per quarter

$i_a = (1.03125)^4 - 1 = .1310$

19) c) $i_m = e^{\frac{.08}{12}} - 1 = .006689$ $\quad , N = 12 \times 20 = 240$.

$F = 100(F|A, .6689\%, 240) = 59{,}098.60$

20) d) Need to finance a \$600,000 house with 20% downpayment:

$600{,}000(1 - .2) = 480{,}000$.

$i_m = \left(1 + \frac{.07}{6}\right)^{\frac{1}{2}} - 1 = .005816, N = 25 \times 12 = 300$

$A = 480{,}000\ (A|P, \overset{.007055}{.5816\%}, 300) = 3386.40$

monthy salary: $S \Rightarrow 0.3 \cdot S \geq 3386.40 \Rightarrow S \geq 11{,}288.00$